

# An IoT-based energy-management platform for industrial facilities



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## HIGHLIGHTS

- Facility energy management systems require interconnection and interoperation.
- An IoT-based communication framework with a common information model is proposed.
- A practical example of an IoT-based energy-management platform.
- Validation of the ability of the platform to enhance system interoperability.
- The platform with improved energy management achieves energy saving targets.

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## ABSTRACT

Interconnectivity and interoperability are very important features in the development of integrated energy management systems for industrial facilities. A simple and common strategy for exchanging energy-related information among the entities in a facility is currently lacking. To this end, the purpose of this study is to present an IoT-based communication framework with a common information model to facilitate the development of a demand response (DR) energy management system for industrial customers. Additionally, we developed and implemented an IoT-based energy-management platform based on a common information model and open communication protocols, which takes advantage of integrated energy supply networks to deploy DR energy management in an industrial facility. The experimental results of this study demonstrate that the proposed platform can not only improve the interconnectivity of the entities in industrial energy management systems but also reduce the energy costs of industrial facilities.

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## 1. Introduction

Industry is the largest consumer of electricity among all end-user sectors. According to statistics from the International Energy Agency, in 2012, the consumption of electricity worldwide by the industrial sector was 42.3% of total energy produced [1]. This has led to significant interest in the development of industrial energy management around the world in recent years [2,3].

The variety of energy management systems, integrated energy systems (IESs) and customer equipment can generate many different data formats, resulting in a lack of interconnectivity and interoperability in industrial energy management systems. The main goal in the integration of energy management systems is to allow the communicating entities to interact with each other using a common information model. This results in the need to be able

to understand and map the meaning and context of information that resides in different domains. In view of this requirement, a common information model is needed to support the ability to semantically align the information for inter-domain communication for the energy management systems in industrial facilities.

The concept of a metamodel for information integration is not new [4], and recently the use of the Extensible Markup Language (XML) as a metadata exchange technology has been widely adopted for providing better information integration between component systems [5,6]. The requirements of semantic interoperability are now being met on the customer side by the use of the Facility Smart Grid Information Model (FSGIM) [7]. The FSGIM is a specification that is currently under development by the American Society of Heating and Air-Conditioning Engineers (ASHRAE). It provides a common data structure that can be used for developing energy management systems in different facility environments. However, currently it does not take into account the specific use of FSGIM in industrial processes. To apply the FSGIM to an

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industrial domain, some extensions and attributes need to be added. In this way, the common information model could be extended to be adaptable for various communication protocols that have already been applied within industrial facilities.

Many industrial communication protocols can be adopted to support FSGIM for industrial energy management. However, applying different private protocols will result in poor interoperability and high development cost. An alternative is the Internet of Things (IoT), an infrastructure of interconnected objects, people, and systems, together with information resources and intelligent services. By using the IoT to interconnect the devices, it is expected to make industrial production more intelligent and efficient. Moreover, IESs and energy management systems for industrial facilities would no longer be stand-alone entities but part of ubiquitous networks. IESs in industrial facilities can be scheduled efficiently and used to their full capacity.

From the viewpoint of energy management schemes, a demand response (DR) program is a key technology that induces users to modify their consumption patterns in response to electricity prices or payment incentives. From the customer side, one of the main purposes of deploying DR is to reduce electricity costs by using more power during times of low prices and less power during times of high prices. Use of DR has been widely studied in commercial and residential facilities, but rarely in industrial. Because manufacturers need to consider not only the overall usage of electricity but also specific resource demands during the operation and production processes, it is difficult to design and implement an IoT-based DR energy management system for industrial consumers. In this study, we focus on designing a common information model and the related IoT-based communication framework for DR management in industrial facilities. The contributions of this paper are as follows:

- The design and implementation of a common information model for an industrial energy management and integrated energy system based on the FSGIM with additional attributes for industrial application.
- The development of an IoT-based communication framework for industrial DR energy management that supports a common information model developed in this work. Under the framework, industrial facilities can easily develop and implement their IoT-based energy management systems. The solution is built on open protocols, such as 6LoWPAN [8], CoAP [9] and IPv6, and provides the possibility to reduce the development cost of an industrial network.
- Development of an IoT-based energy-management platform for an industrial facility, which includes an energy management system (EMS) that runs the DR algorithm, an energy management agent (EMA) that manages industrial tasks, a monitoring and control system (MCS) that monitors and controls the industrial processes, an energy storage system (ESS) that stores energy that can be delivered at a later time, a solar energy generation system (EGS) that generates electrical energy by using the sun's radiation, an industrial Ethernet backbone network to provide reliable communication, and a wireless field network based on 6LoWPAN.
- The experimental results validate the feasibility of the entire IoT-based scheme for an energy management system.

The remainder of this paper is organized as follows. Section 2 discusses related studies. Section 3 introduces the proposed information model and the communication protocol framework. Section 4 introduces the IoT-based energy management system with FSGIM for DR in industrial facilities. Section 5 gives the development of the IoT-based industrial smart energy management platform. Section 6 describes experimental scenarios using the

information model and the IoT communications framework, and also describes an analysis of the test results. Section 7 offers conclusions and possibilities for future work.

## 2. Literature review

Because of the high electricity consumption in industrial applications, it is imperative to establish an industry-centric energy management system as well as to make sure that this system is utilized in a proper way [10]. In the past decade, efforts in academia and industry have shifted toward increasing energy efficiency in manufacturing [11–14]. Lampret [11] implemented an energy-flow management system with energy information center and supervisory control and data acquisition (SCADA) using Ethernet in a pharmaceutical application. Swords et al. [12] introduced a prototype enterprise energy information system that supports strategic energy-management. It provides comprehensive energy monitoring and targeting, integration with energy modeling software and enterprise business databases, and supports measurement and verification. Finn and Fitzpatrick [13] analyzed the potential for the implementation of a price-based DR by an industrial consumer to increase their proportional use of wind-generated electricity. May et al. [14] provided an energy management method for an industrial production application focusing on the means and appropriate performance indicators.

A common information model plays a significant role in achieving the interoperability of energy networks between smart grids and facilities. As defined by the National Institute of Standards and Technology (NIST) [15], the utility meter and the energy service interface (ESI) are placed at the boundary of a facility and exchange communication data between the customer domain and other external domains (e.g., distribution, operation, and market). The other domains in the smart grid provide external energy services to the customer domain. In general, to cope with heterogeneity, a smart grid adopts a canonical data model (CDM) approach [16]. The conceptual model of an industrial facility with a smart grid is shown in Fig. 1. At the grid side, IEC common information model (CIM) [17] families of standards act as the main information model standards. As shown in Fig. 1, open automated demand response (OpenADR) specifications [18,19] can be considered as the smart grid user interface bridge between the grid and the facility. The common information standard used at industrial facilities is the FSGIM, which consists of Load, Meter, EM (Energy Manager) and Generator components (see Section 4.2).

IoT-based protocols can be used within data acquisition and control systems to sense, gather, store, analyze, display and control internal facility processes [20,21]. The important factor is that industrial IoT-based technology should be fully compatible with IP on the communication stack, which provides the possibility to enhance energy management efficiency. There are some studies that have proposed an IoT-based cloud manufacturing service system and its architecture [22,23]. With an IoT-based framework, the number of networked sensors grows dramatically across production, supply chains and products. Shrouf et al. [24] presented a reference architecture for IoT-based smart factories and proposed an approach for energy management based on the IoT paradigm.

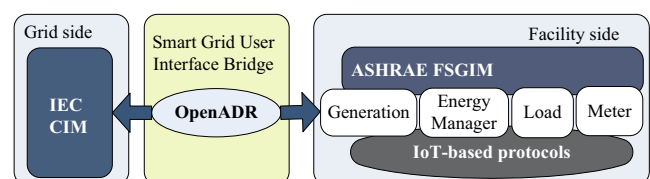


Fig. 1. Smart grid information model standards and relationships between standards.

To enhance energy management efficiency, a great deal of effort has been focused on DR energy management implementations [25–28]. Previous studies proposed a DR energy management model [29] and algorithm [30] for industrial facilities based on a state-task network (STN) [31]. However, most of the existing work focused on DR energy management without considering the integration and interoperability of different systems. To achieve a wide scale DR application, more abstract information models and standardized communication protocols need to be considered. To this end, we propose a common information model and IoT-based communication framework that can enhance the interconnectivity and interoperability of energy management systems.

### 3. The proposed information model and communication framework

Until now, the CIM has been used only to provide interoperability at the grid side. However, the FSGIM is a currently available facility-side specification. The FSGIM defines an abstract, object-oriented information model to enable control systems to manage electrical loads and generation sources in response to communications with an electrical grid. In this paper, the FSGIM is utilized to represent the energy consumption, production, and storage systems in an industrial facility. As a result, a level of interoperability is ensured because the information has been standardized.

Furthermore, the FSGIM defines the data elements, data type, data associations, semantic checks and data optionality. The FSGIM can be used to develop or enhance other standards that define technology and communication protocol specific implementations. It provides the basis for interoperable extensions to the existing communication protocols for facility information. Fig. 2 illustrates the types of information that are standardized in the FSGIM and the relationship between the FSGIM and the communication protocol.

As shown in the middle part of Fig. 2, when applying the FSGIM to industrial energy management systems, development is needed for the communication protocols at different layers, as well as security and additional services. These protocols may use their own existing mechanisms to perform the information encoding and communication. As shown on the right side of Fig. 2, there are several IoT-based communication protocols and solutions that can be used to support FSGIM in industrial energy management systems. IoT-based protocols have several advantages, including an effectively limitless ability to scale, and the ability to accommodate multiple millions of end nodes on a single network.

The purpose of using an IoT-based energy management framework is to allow the system to achieve IP-based remote access through open protocols. Using IoT promotes the ability to exchange energy-related data (collected from ubiquitous devices on the plant floor and from the enterprise energy manager), which is expected to increase cost savings. An IoT communication service with enhanced

communication and sensing capabilities enables interactions between energy producers and end users, and also facilitates the control of distribute energy resources (DERs) and their integration into the main grid.

To implement the FSGIM in industrial facilities, the protocols for both the backbone and wireless networks need to be designed. In this paper, we propose the use of the following communication protocols:

#### (a) Physical and data link layer

In this study, we propose using Industrial Ethernet technology, because it has been widely deployed and has promoted the convergence of plant control and enterprise networks. Additionally, Industrial Ethernet's simple and effective design has made it the most popular networking solution at the physical and data-link levels. Examples of Industrial Ethernet include PROFINET [32], EtherCAT [33], Ethernet Powerlink [34], RAPIEnet [35] and EPA [36]. Energy decision-makers can take advantage of real-time information by accessing key performance indicators and data analytics at the manufacturing application level. The industrial process can be monitored and adjusted in real-time to improve production flexibility.

For the field network, IEEE 802.15.4 [37] is used, which is a standard radio technology for low-power, low-data-rate applications. Because of the ubiquity and the availability of IEEE 802.15.4-compliant radio transceivers, many of the recently developed industrial radio stacks are built on IEEE 802.15.4, including ISA100.11a [38], WirelessHART [39] and WIA-PA [40].

#### (b) Network layer

The opportunity to use open protocols such as IP over Ethernet networks offers the possibility of a level of standardization and interoperability in the industrial field. The transition of IP networks from IPv4 to IPv6 is ongoing. New or improved features, such as transparent end-to-end communication, large addressing space, an auto-addressing method, a more efficient routing protocol, enhanced mobile capability, and autonomous network forming and configuration are very attractive and useful for a plant control network [23]. Plant control networks will migrate to IPv6 to integrate with enterprise intranets and the Internet in the near future.

Besides the IPv6 transition and the integration of different networks, the emergence of IP-enabled wireless field networks is another important trend. The use of IP-based wireless technologies in industrial energy management provides new possibilities and advantages compared to existing wired solutions. These technologies will enable easier access to more information that is related to the process itself and the equipment used in the process.

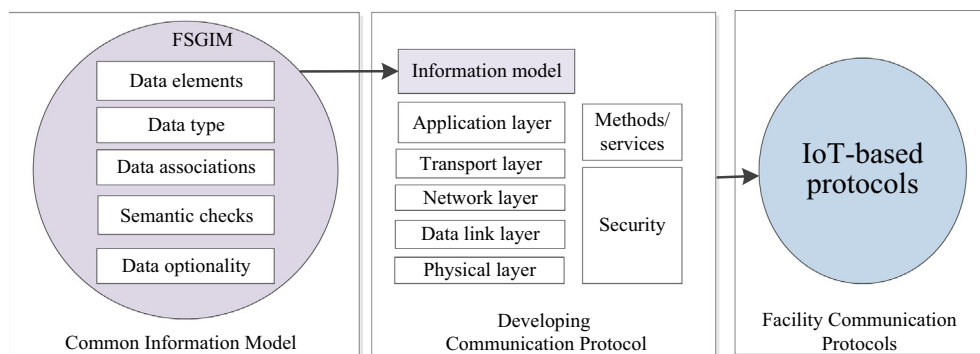


Fig. 2. The relationship between FSGIM and communication protocols.

Lightweight IP stacks and the IPv6-based communication protocol make it possible to enable IP communication in wireless field networks. 6LoWPAN [8,41] is an adapter layer between the IPv6 and IEEE 802.15.4. It is used for low-power and lossy networks (LLNs) where the links used for interconnecting the nodes are IEEE 802.15.4 links. The 6LoWPAN can even be applied to very small devices, including low-power devices with limited processing capabilities allowing them to participate in the IoT. In this work, the energy manager, the load, and the generation systems are located in an industrial facility and are connected by either a wireless or a wired network. The particular communication technology used in the lower layers can be disregarded because the system is based on IP at the network layer.

#### (c) Transport layer

TCP or UDP can be chosen for IoT-based backbone and wireless networks. UDP over 6LoWPAN fits well with many smart object applications. UDP provides a best-effort datagram delivery service, but does not guarantee that the datagrams are delivered to the destination. It is up to the application layer to recover from packet loss. However, the simplicity and lightweight nature of UDP makes it a compelling choice for data that need to be transported quickly such as sensor data.

#### (d) Application layer

For industrial wired part, we chose application layer protocols such as the Hypertext Transfer Protocol (HTTP), for web-style interaction and web service infrastructure, and the Simple Network Management Protocol (SNMP) for network configuration. These allow IP-based smart objects to interoperate with a large number of external systems.

For the wireless part, the application layer protocol also needed to be considered. Constrained Application Protocol (CoAP) [9,42,43] is an application layer protocol that is intended for use in resource-constrained nodes. It is particularly targeted at small low-power sensors, switches, valves and similar components that need to be controlled or supervised remotely through standard Internet networks. CoAP is designed as an easy translation of HTTP for simplified integration with the web, and also meets specialized requirements such as low overhead and simplicity. This makes it possible to use CoAP in one-to-many and many-to-one communication patterns.

In this study, CoAP is used as the application layer protocol for implementing the DR scheme in the wireless part.

#### (e) Web service system

By using web service technology for industrial energy applications, existing web-service-oriented systems, programming libraries, and knowledge can be directly applied. For energy networks, smart energy supply networks and applications can be directly integrated with existing management systems by using the same interfaces. This makes it possible to integrate energy applications into enterprise resource planning systems without any intermediaries, thus reducing the complexity of the system as a whole. For industries, energy management applications can be built using off-the-shelf technology without any customized interfaces or translators.

#### (f) Service discovery

The IP architecture does not have any default service discovery framework. Among the service discovery mechanisms, auto address configuration is especially important, which can be done

either with a centralized protocol such as the Dynamic Host Configuration Protocol (DHCP) or with a distributed mechanism such as IPv4 auto address configuration or IPv6 stateless address configuration. Additionally, some alternative mechanisms such as Service Location Protocol (SLP), Zeroconf and Universal Plug and Play (UPnP) can be used in an industrial energy management system.

#### (g) Security

Security consists of three properties: confidentiality, integrity, and availability. To implement security architecture, encryption can be adopted to convert messages from plaintext into ciphertext, which is not readable by potential attackers. Several other security mechanisms can also be used for energy management systems, such as authentication and key distribution. For computationally constrained smart object microprocessors, hardware-assisted encryption implementations can be adopted to enable strong encryption.

### 4. IoT-based DR energy management system with FSGIM

This section discusses the concept of an IoT-based industrial energy-management platform for DR in industrial facilities. The architecture and the interrelationship among all the model elements are shown in Fig. 3.

#### 4.1. System architecture

The proposed system architecture is divided into the utility side and industrial electricity demand side, with the utility meter as the boundary between the two. As described below, the energy supply networks, i.e., utility power station, energy storage system and energy generation system are integrated with the process tasks as an integrated system.

System elements are essential for building the automatic DR energy-management platform. Each element is uniquely identified by a single symbol, as illustrated in the right box of Fig. 3 [29,30].

- *Production planner/facility manager* is responsible for carrying out production plans. It responds to real-time changes based on feed-back from a process as well as other internal or external events, and is responsible for the maintenance and operation of the facility.
- *Utility power station* acts as an energy supplier and energy information provider. It interacts with the EMS and the smart grid owned Smart Meter.
- *Utility meter* measures energy consumption or generation per time tariff and provides this information to the utility company. Secure communications with this device are provided by the smart grid.
- *Energy Manager System (EMS)* is any device/software or group of the two, installed in an industrial facility that provides the functions of energy management, control and planning in conjunction with responsible facility management. EMS runs the DR algorithm to determine the optimal operating points of tasks and the operating status of IESs to shift the electricity demand from peak to off-peak demand periods, and transmits the control information to MCS. The EMS is located at a top hierarchical level of the industrial network, and can manage and monitor task-level attributes and energy costs of all tasks.
- *Energy Manager Agent (EMA)* monitors the electricity consumption and controls the electric load of each task. The EMA is located at a lower hierarchical level than the EMS in the industrial network, and can manage task-level attributes and energy costs of specific task.



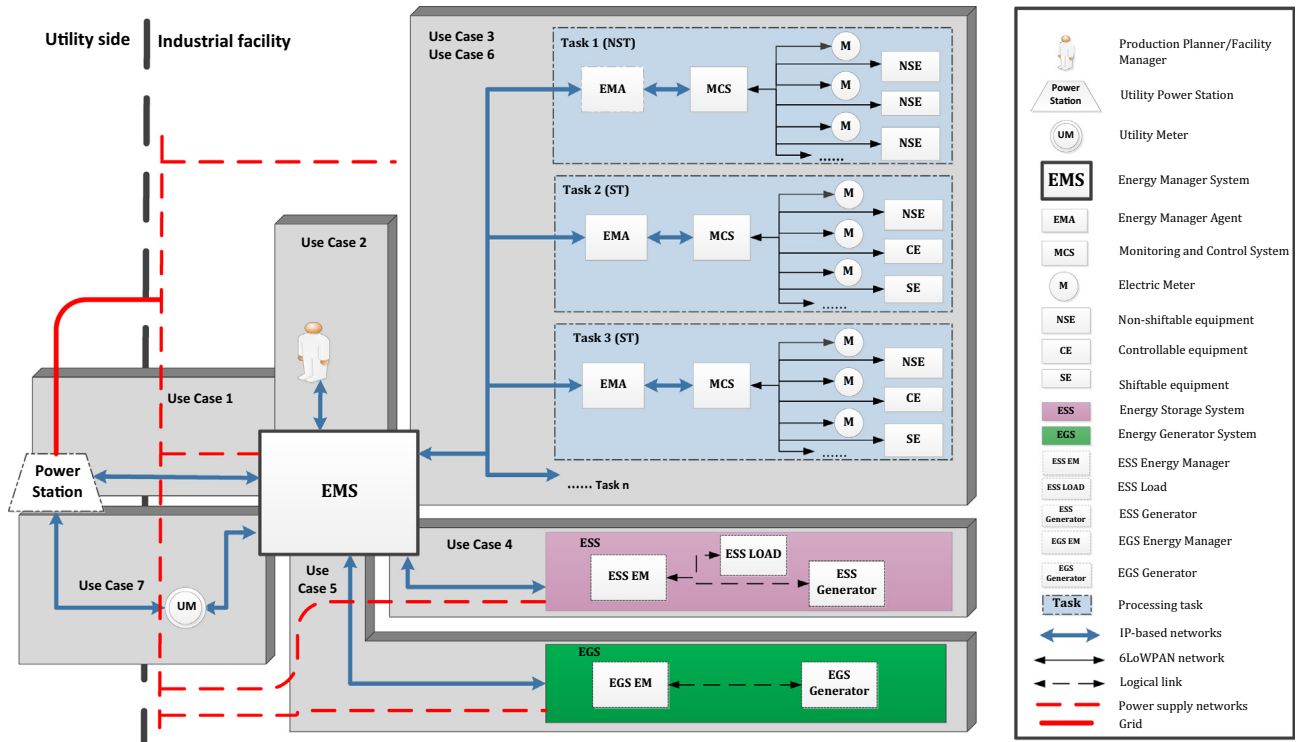


Fig. 3. System architecture.

- **Monitoring and control system (MCS)** is an automation system designed to monitor and control the operation of each device. The MCS is located at the same hierarchical level as the EMA in the industrial network, which can control and monitor load-level attributes and energy costs of specific task.
- **Meter (M)** is a physical device or subsystem onto which an electric meter is defined.
- **Non-shiftable Equipment (NSE)** is a device whose energy demand must be met immediately.
- **Controllable Equipment (CE)** is a device that has multiple operating levels, resulting in differences in electricity demand.
- **Shiftable Equipment (SE)** is a device that can be switched on or off based on the electricity demand.
- **Energy Storage System (ESS)** is a physical device or subsystem that can store electrical energy and whose electrical energy can then be delivered at a later time. For example, a rechargeable battery. ESS is logically divided into ESS EM, ESS Load and ESS Generator.
- **ESS EM** acts as an energy manager performing an internal energy management function to control the electrical storage device.
- **ESS load** acts as a load while the storage device is being charged.
- **ESS generator** acts as a generator while the storage device is supplying electrical energy.
- **Electricity generation system (EGS)** generates electricity in an industrial facility, including solar panels, wind turbines, waste heat recovery, etc. EGS is logically divided into EGS EM and EGS Generator.
- **EGS EM** performs an internal energy management function to control the electricity generation system.
- **EGS generator** acts as a generator while the electricity generation system is supplying electrical energy.

The blue box represents the process task in industrial applications. The red line represents the electric grid and the red dashed

line represents the power supply networks inside the industrial facility. The bold blue arrow line represents the IP-based backbone network and the black arrow line represents 6LoWPAN networks, which were discussed in Section 3. The dashed black arrow line represents the logical links inside ESS and EGS. The seven use cases in Fig. 3 will be explained in Section 4.3.

#### 4.2. Common industrial information model

This section discusses the information models and their instances for this system. As mentioned above, the metadata model in FSGIM XML is a representation of the energy management system proprietary data model and based on open standards that support interoperability. In this study, we consolidate these different metadata representations into a semantically aligned representation of the network reality.

In industrial applications, it is difficult to use the FSGIM to represent all network data parameters. We need to exploit data from industrial applications not previously modeled in the FSGIM, and then semantic alignment is available through harmonization. To do this, we added some additional parameters. This is possible at the metamodel level of abstraction and is referenced in the FSGIM semantic models.

The information model between Power Station and EMS is based on an Energy Market Information Exchange (EMIX) [44], which defines price representations and market interactions. The relationship between the information model and its instance in DR energy management for an industrial facility is shown in Fig. 4. The main common information models that we used in this study include the load model, the meter model, the EM model and the generator model. In Fig. 4, the yellow box represents UM and meter, which are instances of the meter model. The pink box represents equipment, which are instances of the load model. The brown box stands for the energy supply parties, which are instances of the generator model. The green box represents the

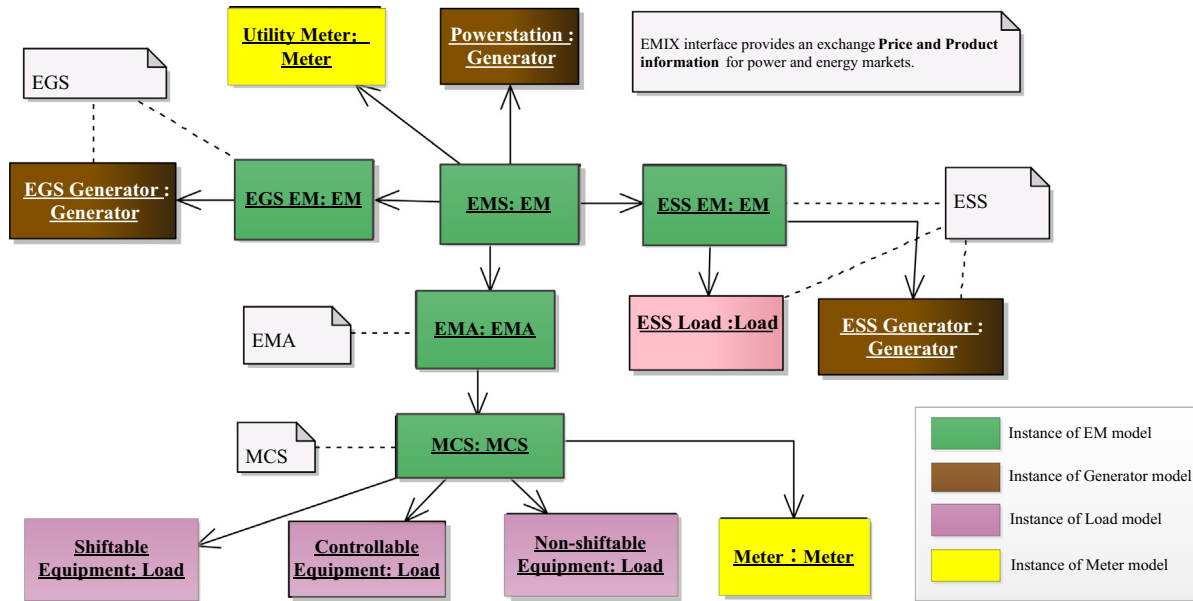


Fig. 4. The relationship between the information models and their instances in DR energy management for industrial facilities.

EM, which are instance of the EM model. We will discuss them as follows:

#### (a) Load model

The *Load* component defines the common attributes to all devices that consume electricity. In the DR model, processing tasks are divided into non-schedulable tasks (NSTs) and schedulable tasks (STs), and both are composed of pieces of industrial equipment. NSTs are tasks for which the demand cannot be scheduled, and must be satisfied immediately despite the fact that the price of electricity is high or low. STs are tasks for which the demand can be scheduled among a pre-specified set of operating points [30].

The industrial equipment is classified as NSEs, SEs and CE. NSTs must contain NSEs only, and STs consist of NESs, SEs and CEs. The FSGIM *Load* component is used directly to represent NSEs, SEs and CEs.

In the FSGIM, *CurtableLoad* is a child class of *Load*. The *CurtableLoad* class defines attributes that are unique to devices whose electrical consumption can be curtailed. The attributes define the information that is needed to represent curtailment policies based on criteria including load priorities, cost constraints, protection of physical equipment, and safety considerations. The *CurtailmentRatingsLevel* is an attribute of *CurtableLoad*. This class defines a single array. Each array element defines the fixed demand to which the load shall be controlled as a function of the stated level. The *Priority* is an attribute of *CurtableLoad*, which defines the curtailment priority and the order upon which curtailment occurs across a set of loads. A *Priority* value of 1 is considered a critical load that may not be shed except during a critical event.

- CE is modeled as an instance of the *CurtableLoad* class model.
  - SE is modeled as an instance of the *CurtableLoad* class with one curtailment ratings level.
  - NSE is modeled as an instance of the *CurtableLoad* class model with priority equal to 1, i.e., the highest priority.
- (b) Meter model

A meter is modeled as an instance of the *Meter* class in the FSGIM. We use the FSGIM *Meter* component directly, which pro-

vides an abstract representation for the function of measuring power, energy, and emissions. The instance of meter is a physical device or subsystem which measures the demand or supply of one or more loads and/or one or more generators.

#### (c) Energy manager (EM) model

The FSGIM defines the EM component to provide an abstract representation of energy management system functionality. The functionality is implemented in any device that performs analysis and makes energy-related decisions involving meters, loads, generators, or energy storage devices. In this platform, EMS, EMA and MCS are instances of the EM model.

In view of this situation, it is difficult to use the EM model that is defined in the FSGIM to represent the industrial STN elements completely, which allows for development of a general algorithm to find an optimal solution for the factory energy management system (see [30]). The EMS, EMA and MCS models need to be designed by adding some classes based on the original FSGIM EM model. The main attributes of EMS, EMA and MCS are shown in Table 1.

- EMS model: The EMS model includes the necessary attributes for managing and monitoring the electricity demand of whole industrial facilities. As a result, it can maintain the task attributes and operation information of whole industrial applications. The EMS as the instance of the EMS model establishes communications with the utility suppliers via a wide area network (WAN) to obtain the day-ahead dynamic electricity prices. The EMS transmits control commands to each MCS via the industrial Ethernet backbone network.
- EMA model: The EMA model includes the necessary attributes for managing and monitoring energy for a task. As a result, it can maintain the task attributes and operation information of the specific task.
- MCS model: The MCS model includes the necessary attributes for controlling and measuring the energy consumption of each processing task. As a result, it maintains the task attributes and load operation point attributes of a specific task. It controls the field devices to operate at specific operating points requested by the EMS by sending control commands.

#### (d) Generator model

**Table 1**  
EM model class attributes.

| Model | Attribute name                | Description                                                                                                                   |
|-------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| EMS   | TaskNumber                    | This attribute defines the number of tasks that are managed by the EMS                                                        |
|       | SetTasks                      | This attribute defines the set of tasks, including the NSTs and STs                                                           |
|       | SetStates                     | This attribute defines the set of states, which represent the related information of feeds, intermediates, and final products |
|       | AllNodeNumber                 | This attribute defines the number of nodes that are managed by the EMS                                                        |
|       | SupportedOperationInformation | This attribute defines the supported operation information of each task                                                       |
| EMA   | EMSOOperationInformation      | This attribute defines the historical, current and next operation point information                                           |
|       | TaskID                        | This attribute defines the task ID that is managed by the EMA                                                                 |
|       | TaskType                      | This attribute defines the types of the related task                                                                          |
|       | TaskNodeNumber                | This attribute defines the number of nodes that are managed by the EMA                                                        |
|       | EMAOperationInformation       | This attribute defines the operation information of the related task                                                          |
| MCS   | ArrayOperationPoint           | The attribute defines the array of the operation point and load operation level                                               |
|       | TaskID                        | This attribute defines the task ID that is managed by the MCS                                                                 |
|       | TaskType                      | This attribute defines the type of the related task                                                                           |
|       | TaskNodeNumber                | This attribute defines the number of nodes that are managed by the MCS                                                        |
|       | MCSOperationInformation       | This attribute defines the operation information of the related task                                                          |
|       | ArrayOperationPoint           | The attribute defines the array of the operation point and load operation level                                               |
|       | CurrentElectricityDemand      | This attribute defines the current electricity demand                                                                         |

We use the FSGIM *Generator* component directly. Many classes of the *Generator* component are derived from the IEC 61850-7-420 [45] standard. The *Generator* component provides an abstract representation of all devices that produce electricity or store it. In our platform, the generation element of EGS and ESS can be considered as an instance of the *Generator* model.

The ESS *Generator* is an instance of the *Generator* model while the storage device is providing electrical energy. The EGS *Generator* is also an instance of the *Generator* model while the EGS is supplying electrical energy. A power station can also be considered as an instance of the *Generator* model.

#### 4.3. Use cases

Several DR-related case studies have been conducted to examine the availability of the IoT-based smart grid platform with a common information model for industrial facilities. The use cases are presented in this section.

- *Use Case 1: Determining energy/demand price information*

In this use case, the power station provides the dynamic pricing to the facility. These price data are developed by the power station using internal procedures to maintain a balance between generation and supply near the time of use. Typically these price data are used by facilities to manage their current operating consumption. The price data are used by the facility control systems to control resources, subject to ensuring that all production safety, performance and product quality requirements are maintained. The communication between the utility power station and the EMS is based on using a wide area network (WAN), in which TCP/IP can guarantee reliability of packet transfer.

- *Use Case 2: Determining DR parameters*

In this use case, the EMS prepares the parameters of the DR algorithm and makes the DR decision. The DR algorithm is formulated using the STN model and mixed integer linear programming (MILP) [30]. These parameters include the STN model parameters, the supported operating point information of each task, the operating parameters of the EGS and the operating parameters of the ESS.

Based on the inputs including the day-head electricity price, the STN representation of industrial facilities, the operating information of each task, and the operating information of DERs, the DR algorithm selects the optimal operating point of STs for each pre-

specified time interval. By scheduling STs with different operating points, the DR algorithm can shift part of the electricity demand from peak to off-peak demand periods.

The DR algorithm also determines the optimal power source (i.e., the grid, EGSs, or ESSs) for each time interval. For example, during off-peak periods, the electrical grid supplies electricity to industrial facilities, and the ESS charges energy from the power grid, whereas during peak periods, the EGS (which may be solar, wind, or waste heat from power plants) supplies electricity, and the ESS discharges to supply energy to industrial facilities. The detailed DR algorithm for industrial facilities is described in [30].

- *Use Case 3: Managing the operation point of each time interval to minimize cost*

In this use case, the facility manages the operating point of each time interval to minimize cost. After making a DR decision, the EMS provides the EMA of each task, the optimal operating point of the task and the operating time of that operating point. The EMA proposes operating levels of the equipment in each task and sends this information to the MCS, which finally controls the equipment according to the commands scheduled by the EMA.

When the state of the load changes after receiving a control message, the wireless equipment relays the updated state to the MCS and EMA. The EMA sends the task level response to the EMS. These response messages not only act as the acknowledgment but also carry the important attributes.

- *Use Case 4: Determining the utilization of ESS*

In this use case, the facility decides to buy power or utilize the ESS within each time interval. After determining the energy price information and the DR parameters, the EMS specifies the ESS operating mode in the next stage and the operating duration of that mode, which is an IES decision. The communication between the EMS and ESS is based on TCP, which guarantees reliability of packet transfers.

- *Use Case 5: Determining the utilization EGS*

In this use case, the facility decides to buy power from the grid or use the EGS within each time interval. After making a DR decision, the EMS specifies the EGS operating mode in the next stage and the operating duration of that mode, which is an IES decision. In a high-electricity-price period, it encourages the EMS to com-

mand the EGS to supply electricity to some or all of the processing tasks. This decreases the electricity demand of the industrial facility.

- *Use Case 6: Measuring equipment power consumption*

In this use case, the facility measures equipment power consumption for each task and each load. The facility measures the power consumption of a particular electrical device and each task.

Prior to its use, the operation manager installs a meter to measure energy at a device or for each task. Periodically, the meter measures the energy consumption of each device and sends an energy measurement message to the MCS to track the ongoing energy use of the load. The MCS sends the energy consumption of related loads to the EMA. The EMA sends the energy consumption of each task to the EMS. The EMS supplies the energy consumption to the facility manager.

- *Use Case 7: Measuring all energy consumption in a factory*

In this use case, the facility measures all equipment power and provides this information to the utility power station and the EMS. The utility meter measures the whole energy consumption of the factory. The utility meter sends the energy measurement message to the EMS to track the ongoing energy use of the factory. The EMS provides the energy consumption information to the facility manager. The Utility Meter sends the energy measurement message to the Utility Power Station, which is used by an energy provider to track facility or equipment performance and analysis.

## 5. Development of the experimental platform

In this section, we introduce the integrated hardware and software experimental platform that we developed in this study. In Fig. 5, the software simulation components are shown on left-hand side, and the real hardware components for the implementation are shown on right-hand side. In our previous work [46], a hardware-in-the-loop system for demand response energy management in industrial facilities was implemented using an ISA100.11a system and RAPIenet network. The solution was based on private protocols with high expenses and the data structure was not based on a common information model; as a result the interactions and interconnections for the energy management networks were extremely complex.

In this study, a new industrial energy-management platform has been prototyped and developed, as shown in Fig. 5. We implemented IP-based Ethernet as the backbone network. Meanwhile, we implemented 6LoWPAN over IEEE 802.15.4 as the wireless part. Here, the field node applications can be directly integrated with existing IP-based systems and use the standardized interfaces, which drastically reduces the complexity of the system as a whole. Moreover, we implemented HTTP for web-style interactions in the wired part and implemented CoAP for DR energy management in the wireless part. For the web services, the browser/server (B/S) software architecture has been implemented to replace the previous client/server (C/S) architecture in our work, which was able to meet the current global network of open and interconnected devices. Moreover, the production planner or facility manager can access the EMS, EMA and MCS by using these web services.

This proposed system can accommodate many types of DERs including solar panels, wind turbines and wasted heat recovery generated in the facility. Here, we only include a simulation model of a solar EGS in the experimental scenario because the size of the experimental model is small and the power consumption is limited. The structure of the system was composed of the simulation component and the physical component.

### 5.1. The software simulation

We used a software simulation to realize the PowerStation, the EGS and the ESS in energy management system.

#### (a) PowerStation

The PowerStation was realized by using a high-performance personal computer, on which a virtual utility supplier was built. It periodically provided day-ahead dynamic electricity prices to the EMS.

#### (b) ESS

A virtual ESS was developed as an in-facility DER using a high-performance personal computer (PC), which was developed using a software model. Here, the energy manager part of the ESS performs an internal energy management function to control the electrical storage device, which follows the EM model defined in the FSGIM.

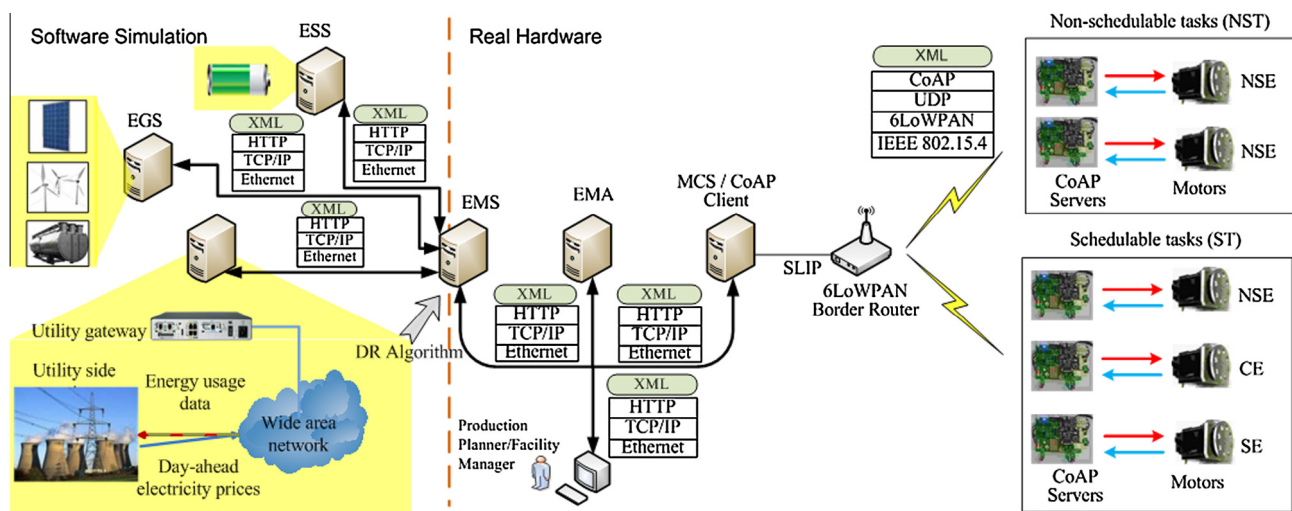


Fig. 5. An overview of the energy management platform for industrial facilities.



## (c) EGS

A virtual in-facility solar EGS was also developed using a software model on the PC. According to the FSGIM, energy is generated by the EGS Generator model, whereas the energy management function is performed by the EGS EM model for the solar EGS.

## 5.2. The physical component

This section describes the physical component of the developed platform.

## (a) EMS

In this study, the EMS was developed using a PC. Based on day-ahead electricity prices, the EMS runs the DR algorithm to determine the optimal operating points of the STs and the operating status of the IESs to shift the electricity demand from peak to off-peak demand periods, and transmits the control information to the EMA and MCS.

## (b) EMA

In this study, the EMA was developed using a PC. The EMA is acted as the energy manager agent for monitoring the electricity consumption and controls the electric load of each task.

## (c) MCS

In this study, the MCS was built using a PC, which communicates with the EMA and EMS using Ethernet as a backbone network. A 6LoWPAN border router was also connected with the MCS using the Serial Line Internet Protocol (SLIP) [47,48] for enabling communication between the backbone network and the wireless field network. In the wired part, the communication is based on HTTP over TCP, whereas in the wireless part, communication is based on COAP over UDP. Therefore, a mapping function for wireless and wired protocols was implemented in the MCS.

## (d) Wireless field networks

In our experimental facility, we used 6LoWPAN as the wireless communication network. In each field device, we used DC motors

as industrial loads, because motors are one of the most commonly used pieces of industrial equipment. Fig. 5 shows that the industrial process side included two tasks: NST and ST. The NST consisted of two motors, which were regarded as NSEs, and the ST consisted of three motors, which were regarded as representing each of the three categories (NSE, SE, and CE).

## 5.3. The Web-service implementation with XML based on FSGIM

The Web service is a mechanism for exchanging data between different systems developed by different parties. In this study, the Web-service implementation made the exchange data in a system-independent way. Here, FSGIM was implemented to standardize the data formats. XML was used to specify the FSGIM as a metadata exchange technology. HTTP/CoAP used XML as their data format. As a general purpose document format, XML provides a structured mechanism to encode machine-readable information.

The structure and semantics of the FSGIM are defined using the Unified Modeling Language (UML). UML diagrammatic conventions can define the normative relationships and multiplicity details of the FSGIM [7].

As discussed in Section 4.2-c) (see Table 1), additional attributes and relationships were added for the EMS, EMA and MCS in the industrial facilities. Then we made the updated FSGIM UML file using the Enterprise Architect (EA) tool. For each component, the EA tool could convert the UML class to XML schema. An XML schema is a description of a type of XML document, typically expressed in terms of constraints on the structure and content of documents of that type, above and beyond the basic syntactical constraints imposed by XML. XML is an instance of an XML schema. Each device may have more than one XML file. Fig. 6 gives an example of an XML schema for the meter.

On the wireless side, the limited resources in field nodes regarding processing capacity, power supply, and communication bandwidth necessitated the use of lightweight mechanisms. Representational state transfer (REST) is a lightweight instantiation of the Web services concept, which is particularly well suited to the properties of smart objects. In this system, the RESTful architecture was implemented. With this instantiation of REST, CoAP requests were used to transfer representations of resources between clients and servers.

```
<xs:include/>>
<xs:element name="ElectricMeter" type="ElectricMeter"/>
<xs:complexType name="ElectricMeter">
  <xs:annotation>
    <xs:documentation>The ElectricMeter Class is a subclass of the Meter Class. It is an abstract representation
  </xs:annotation>
  <xs:complexContent>
    <xs:extension base="Meter">
      <xs:sequence>
        <xs:element name="powerQuality" type="ElectricPowerQualitySummary" minOccurs="0" maxOccurs="1">
          <xs:annotation>
            <xs:documentation>The power quality statistical data set for a given location.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="input" type="ConnectionPoint" minOccurs="1" maxOccurs="1"/>
        <xs:element name="output" type="ConnectionPoint" minOccurs="1" maxOccurs="1"/>
        <xs:element name="powerReadings" type="PowerMeasurementsSet" minOccurs="0" maxOccurs="1">
          <xs:annotation>
            <xs:documentation>The power measurement set for a given location.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="powerReadings" type="PowerMeasurementsSet" minOccurs="0" maxOccurs="1">
          <xs:annotation>
            <xs:documentation>The power measurement set for a given location.</xs:documentation>
          </xs:annotation>
        </xs:element>
        <xs:element name="energyReadings" type="EnergyMeasurementsSet" minOccurs="0" maxOccurs="1">
          <xs:annotation>
            <xs:documentation>The energy measurement set for a given location.</xs:documentation>
          </xs:annotation>
        </xs:element>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
```

Fig. 6. XML schema based on FSGIM.

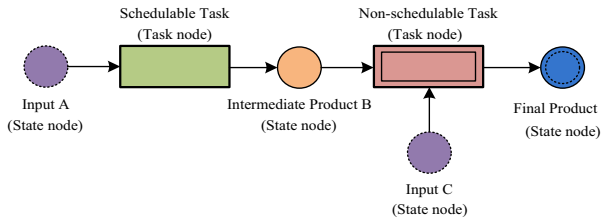


Fig. 7. The STN model of the experimental scenarios.

## 6. Experimental scenarios and results

### 6.1. Experimental scenarios

As shown in Fig. 7, we developed an STN model of the energy-management platform, which included four state nodes (input A, input C, intermediate product B and final product D) and two task nodes (schedulable task and non-schedulable task) (see [46] for a more detailed description of this experimental scenario). In this model, the ST consumes input A to produce intermediate product B, and the NST consumes intermediate product B and input C to produce final product D.

Tables 2 and 3 list the parameters of the NSTs and STs, respectively. The NSTs have only one operating point, specified by parameters such as the speed of each motor, power consumption, and the consumption and production rate of the related state nodes. The STs have five operating points, with the same parameters defined for each operating point.

In the experimental scenarios, the storage for input A, intermediate product B, and input C were modeled using software. The storage capacity and initial storage of intermediate product B were set to 500 and 50, respectively (Table 4).

In the experimental scenario, IESs were considered by using an ESS and a solar EGS, both present inside the facility. For the ESS parameters, initial storage was 0 mW h, storage capacity was assumed to be 2500 mW h, maximum charging and discharging rates were set to 600 mW, and the charging and discharging efficiency coefficients were set to 0.9 (Table 4). It was assumed that the rate of electricity charged/discharged during a given time interval could be continuously controlled between zero and the maximum charging/discharging rate.

There can be many different forms of facility energy generation system, such as solar, wind turbine and wasted heat recovery. In this experimental scenario, a solar EGS is used for software simulation. Fig. 8 shows the predicted energy generated by the solar EGS during each time interval. The energy generation rate was high

Table 4  
Parameters of Intermediate Product B and ESS.

| Parameter  | Description                           | Value     |
|------------|---------------------------------------|-----------|
| B_Cap      | Intermediate product storage capacity | 500 units |
| B_Init     | Intermediate product initial storage  | 50 units  |
| $I_{ess}$  | ESS initial storage                   | 0 mW h    |
| $C_{ess}$  | ESS storage capacity                  | 2500 mW h |
| $\theta_c$ | ESS maximum charging rate             | 600 mW    |
| $\theta_d$ | ESS maximum discharging rate          | 600 mW    |
| $\gamma_c$ | ESS charging efficiency               | 0.9       |
| $\gamma_d$ | ESS discharging efficiency            | 0.9       |

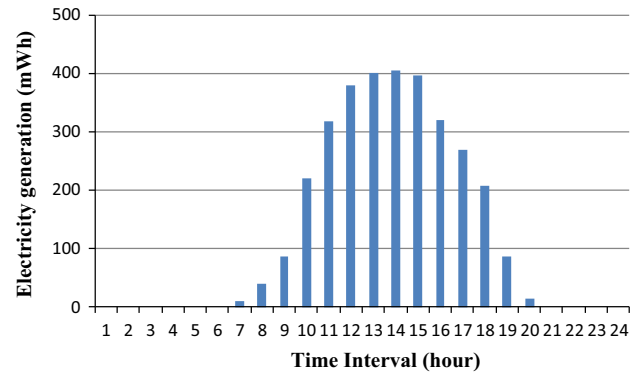


Fig. 8. The predicted quantity of electricity generated by the solar EGS during each time interval.

during the peak sun hours, while no energy was generated through the night.

### 6.2. Evaluation of the implementation

A wireless field network was implemented using a CC2538 microprocessor with the Contiki operating system. Contiki provides effective low-power communication ability and supports fully standard IPv6 and IPv4, as well as 6LoWPAN [49]. The Contiki operating system has also been demonstrated as an easy-to-use platform for building CoAP-enabled DR energy management.

The CoAP resources provided on the meter and the load are presented in Table 5. For the meter, a CoAP GET request, which is issued by the CoAP client, retrieves energy consumption from the meter. The CoAP server, which runs on the meter, sends the CoAP response with the energy consumption using XML structure to the CoAP client. For SE and CE of the load instances, a CoAP PUT

Table 2  
Parameters of devices for non-schedulable task.

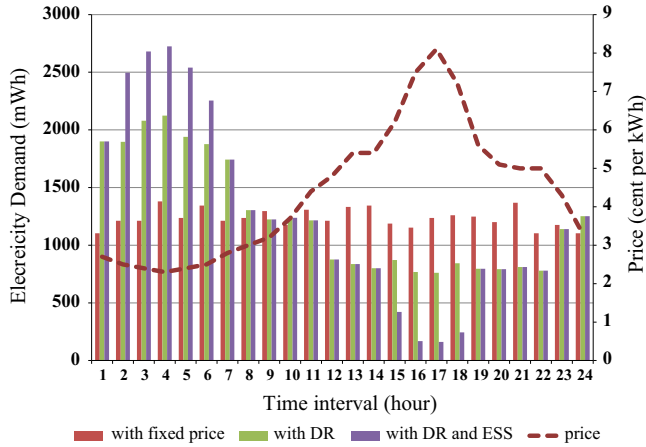
| Operating point | Non-shiftable device (motor 1) rpm | Non-shiftable device (motor 2) rpm | Power consumption (mW h) | Consumption rate of B (unit) | Consumption rate of C (unit) | Production rate of D (unit) |
|-----------------|------------------------------------|------------------------------------|--------------------------|------------------------------|------------------------------|-----------------------------|
| 1               | 1200                               | 600                                | 620                      | 60                           | 120                          | 100                         |

Table 3  
Parameters of devices for schedulable task at each operating point.

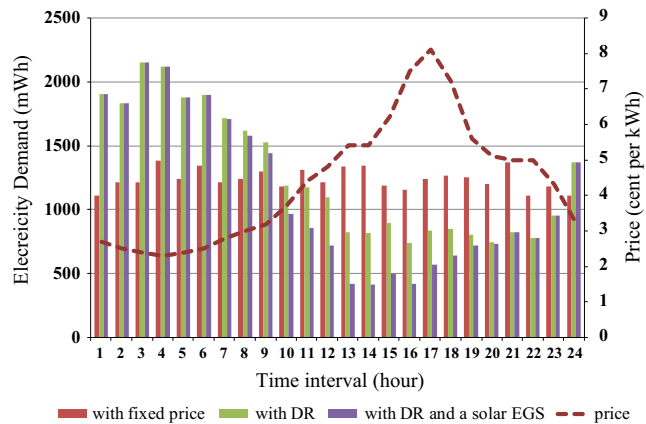
| Operating point | Non-shiftable device (motor 3) rpm | Controllable device (motor 4) rpm | Shiftable device (motor 5) rpm | Power consumption (mW h) | Consumption rate of A (unit) | Production rate of B (unit) |
|-----------------|------------------------------------|-----------------------------------|--------------------------------|--------------------------|------------------------------|-----------------------------|
| 1               | 600                                | 0                                 | 0                              | 210                      | 30                           | 20                          |
| 2               | 600                                | 600                               | 0                              | 420                      | 60                           | 40                          |
| 3               | 600                                | 1200                              | 0                              | 635                      | 90                           | 60                          |
| 4               | 600                                | 1800                              | 0                              | 940                      | 120                          | 80                          |
| 5               | 600                                | 1800                              | 1200                           | 1365                     | 150                          | 120                         |

**Table 5**  
CoAP resources on field wireless nodes.

| Resource   | GET | PUT | Description                      |
|------------|-----|-----|----------------------------------|
| /es        | X   |     | Energy consumption               |
| /loadlevel | X   | X   | The load CurtailmentRatingsLevel |



**Fig. 9.** Experimental results energy consumption and electricity price with DR and ESS included.

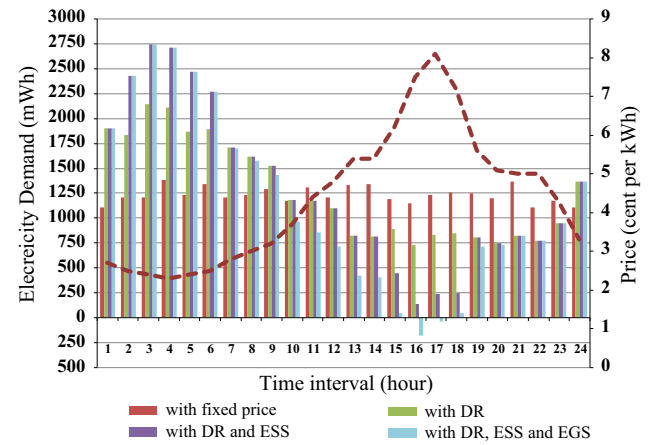


**Fig. 10.** Experimental results of energy consumption and electricity price with DR and EGS included.

request, issued by the CoAP client, sets the curtailment rating level attribute of the load, based on which the CoAP server controls the operating status of the load. The *CurtailmentRatingsLevel* attribute can also receive a CoAP GET request from the CoAP client with XML structure.

According to the dynamic change of price in each time interval, NST operated at the same operating point, while ST operated at different operating points. When the price was low, ST was scheduled to operate at a high energy consumption operating point, thereby increasing demand. When the price was high, ST was scheduled to operate at a low energy consumption operating point, thereby decreasing demand.

Fig. 9 shows the energy demand under three different scenarios. In the fixed-price scenario, the electricity price was constant over all time intervals and was set to be equal to the average of the dynamic price. Compared with the fixed-price scenario, the scenario with DR consumed more electrical energy during low-price periods and less electrical energy during high-price periods, and thus the total cost was reduced. In the case of the scenario with



**Fig. 11.** Experimental results of energy consumption and electricity price with DR, ESS and EGS included.

**Table 6**  
Total energy costs of experimental scenarios.

|                 | Fixed price | DR price | DR price with ESS | DR price with EGS | DR price with ESS and EGS |
|-----------------|-------------|----------|-------------------|-------------------|---------------------------|
| Costs (cent \$) | 1293.01     | 1136.69  | 1039.03           | 979.76            | 898.94                    |

DR and ESS, the gap in energy demand between its maximum and minimum becomes larger. Among the three scenarios, because ESS stored energy when prices were low and supplied energy when prices were high, the scenario with DR and ESS consumed the most electrical energy during low-price periods and the least electrical energy during high-price periods. With DR and ESS, the total cost was reduced even further, not only by shifting demand but also by managing IESs. The total electrical energy costs with DR and ESS decreased by 19.64% in comparison to the fixed-price scenario.

Fig. 10 shows the energy demand under three additional scenarios. In the case of the scenario including DR and EGS, the electricity demand decreased further because the solar EGS generated electricity during time intervals 7–19. The EGS reduced energy costs by generating additional electricity using solar power. The total electrical energy costs when DR and EGS were included decreased by 24.22% in comparison to the fixed-price scenario.

Fig. 11 shows the energy demand under a further four different scenarios. In the case of the scenario with DR, ESS and EGS, the ESS reduced energy costs because it stored electricity when prices were low for use when prices were high, and the EGS reduced energy costs by generating additional electricity using solar power. The electricity demand was negative during time intervals 16–17, which indicates that the facilities may sell surplus electricity to the grid to make a profit. The total electrical energy costs when DR, ESS and EGS were included decreased by 30.48% in comparison to the fixed-price scenario.

Table 6 shows the total energy cost for different cases (fixed price corresponds to the electricity price in each time interval being fixed, and equal to the average of the dynamic price).

## 7. Conclusion and future work

This paper focuses on proposing an IoT-based communication framework with a standard common information model. The application of the FSGIM to IES in industrial facilities is discussed. Current results show that the IoT-based architecture and FSGIM based information models can play an important role in providing interconnectivity and interoperability between devices and equipment,

which guarantees simpler implementation, installation, operation and management of integrated energy management systems in a facility. It also offers a simple and common strategy for exchanging energy-related information between the entities in a facility.

In this study, we developed an IoT-based energy-management platform with common information models for industrial facilities. The integrated energy system architecture and related cases for utilizing different types of energy (EGS, ESS and utility side) are discussed. The IoT-based energy management system includes EMS, EMA, MCS, an industrial Ethernet backbone network and a wireless field network.

The results demonstrate that energy demand could be shifted from peak to off-peak demand periods, leading to significant reductions in overall cost. Additionally, the IoT-based technologies allowed the system to be implemented with less time, effort, complexity, and cost.

Future work will extend and implement the DR scheme to support the real-time energy price in the Cyber Physical System (CPS) [50–52] – IoT platform. The IoT-based energy-management platform will be extended by developing the actual implementation of IESs, such as real rechargeable batteries, solar panels, wind turbines and wasted heat recovery. We also plan to extend the application to real industrial processes.

## Acknowledgements

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## References

- [1] International Energy Agency. Key world energy statistics; 2014.
- [2] Abdelaziz EA, Saidur R, Mekhilef S. A review on energy saving strategies in industrial sector. *Renew Sustain Energy Rev* 2011;15(1):150–68.
- [3] Trianni A, Cagno E, Farné S. Barriers, drivers and decision-making process for industrial energy efficiency: a broad study among manufacturing small and medium-sized enterprises. *Appl Energy* 2016;162(15):1537–51.
- [4] Hargreaves N, Taylor G, Carter A, McMorran A. Developing emerging standards for power system data exchange to enable interoperable and scalable operational modelling and analysis. In: Proceedings of 46th international universities' power engineering conference (UPEC); 2011. p. 1–5.
- [5] Ferreira R, Moura-Pires J, Martins R, Pantoquillo M. XML based metadata repository for information systems. In: Proceedings of artificial intelligence portuguese conference. IEEE; 2005. p. 205–13.
- [6] Rueda C, Bermudez L, Fredericks J. The MMI ontology registry and repository: a portal for marine metadata interoperability. In: Proceedings of OCEANS 2009, MTS/IEEE biloxi-marine technology for our future: global and local challenges. IEEE; 2009. p. 1–6.
- [7] American Society of Heating and Air-Conditioning Engineers (ASHRAE)/National Electrical Manufacturers Association (NEMA) SPC 201 Facility Smart Grid Information Model (FSGIM); 2014.
- [8] Kushalnagar N, Montenegro G, Schumacher C. IPv6 over low-power wireless personal area networks (6LoWPANs): overview, assumptions, problem statement, and goals. RFC4919; 2007.
- [9] Shelby Z, Hartke K, Bormann C. The constrained application protocol (CoAP). RFC 2014;7252.
- [10] Rudberg M, Waldemarsson M, Lidestam H. Strategic perspectives on energy management: a case study in the process industry. *Appl Energy* 2013;104:487–96.
- [11] Lampret M, Bukovec V, Paternost A, Krizman S, Lojk V, Golobic I. Industrial energy-flow management. *Appl Energy* 2007;84(7):781–94.
- [12] Swords B, Coyle E, Norton B. An enterprise energy-information system. *Appl Energy* 2008;85(1):61–9.
- [13] Finn P, Fitzpatrick C. Demand side management of industrial electricity consumption: promoting the use of renewable energy through real-time pricing. *Appl Energy* 2014;113:11–21.
- [14] May G, Barletta I, Stahl B, Taisch M. Energy management in production: a novel method to develop key performance indicators for improving energy efficiency. *Appl Energy* 2015;149:46–61.
- [15] NIST Framework and Roadmap for Smart Grid Interoperability Standards, Release 2.0; 2012.
- [16] Semantic Model Working Party, Smart Grid Architecture Committee. <<http://collaborate.nist.gov/twiki-sggrid/bin/view/SmartGrid/SGIPSemanticModelSGAC>>.
- [17] Usilar M, Specht M, Rohjans S, Trefke J, González M. The common information model CIM: IEC 61968/61970 and 62325-A practical introduction to the CIM. Springer Science & Business Media; 2012.
- [18] Alliance, OpenADR. OpenADR 2.0 profile specification; 2011.
- [19] IEC 62746-10-1 Ed. 1.0 Systems interface between customer energy management system and the power management system – Part 10-1: open automated demand response (OpenADR 2.0b Profile Specification); 2015.
- [20] Tao F, Ying Z, Xu LD, Lv L. Internet of things and BOM based life cycle assessment of energy-saving and emission-reduction of product. *Ind Inform, IEEE Trans* 2014;10(2):1252–61.
- [21] Bi Z, Da Xu L, Wang C. Internet of things for enterprise systems of modern manufacturing. *Ind Inform, IEEE Trans* 2014;10(2):1537–46.
- [22] Tao F, Cheng Y, Xu LD, Zhang L, Li BH. CCloud-CMfg: cloud computing and Internet of things-based cloud manufacturing service system. *Ind Inform, IEEE Trans* 2014;10(2):1435–42.
- [23] Vasseur JP, Adam D. Interconnecting smart objects with IP: the next Internet. Morgan Kaufmann; 2010.
- [24] Shrouf F, Ordieres J, Miragliotta G. Smart factories in industry 4.0: a review of the concept and of energy management approached in production based on the Internet of things paradigm. In: Proceedings of IEEE international conference on industrial engineering and engineering management (IEEM). IEEE; 2014. p. 697–701.
- [25] Siano P. Demand response and smart grids – a survey. *Renew Sustain Energy Rev* 2014;30:461–78.
- [26] Marzband M, Sumper A, Domínguez-García JL, Gumara-Ferret R. Experimental validation of a real time energy management system for microgrids in islanded mode using a local day-ahead electricity market and MINLP. *Energy Convers Manage* 2013;76:314–22.
- [27] Marzband M, Ghadimi M, Sumper A, Domínguez-García JL. Experimental validation of a real-time energy management system using multi-period gravitational search algorithm for microgrids in islanded mode. *Appl Energy* 2014;128:164–74.
- [28] Neves D, Pina A, Silva CA. Demand response modeling: a comparison between tools. *Appl Energy* 2015;146:288–97.
- [29] Ding Y, Hong SH. A model of demand response energy management system in industrial facilities. In: Proceedings of IEEE international conference on smart grid communications; 2013. p. 241–6.
- [30] Ding Y, Hong SH, Li X. A demand response energy management scheme for industrial facilities in smart grid. *Ind Inform, IEEE Trans* 2014;10(4):2257–69.
- [31] Kondili E, Pantelides CC, Sargent RWH. A general algorithm for short-term scheduling of batch operations-I. MILP formulation. *Comput Chem Eng* 1993;17(2):211–27.
- [32] Ferrari P, Flammini A, Vitturi S. Performance analysis of PROFINET networks. *Computer Standards Interf* 2006;28(4):369–85.
- [33] Jansen D, Buttner H. Real-time Ethernet: the EtherCAT solution. *Comput Control Eng* 2004;15(1):16–21.
- [34] Erwinski K, Paprocki M, Grzesiak LM, Karwowski K, Wawrzak A. Application of Ethernet Powerlink for communication in a Linux RTAI open CNC system. *Ind Electron, IEEE Trans* 2013;60(2):628–36.
- [35] Abbas SH, Hong SH. A scheduling and synchronization technique for RAPIenet switches using edge-coloring of conflict multigraphs. *Commun Networks J* 2013;15(3):321–8.
- [36] Lu L, Feng D, Chu J. Improving the real-time performance of Ethernet for plant automation (EPA) based industrial networks. *J Zhejiang Univ Sci C* 2013;14(6):433–48.
- [37] IEEE 802.15.4-2011 – IEEE Standard for Local and metropolitan area networks – Part 15.4: Low-Rate Wireless Personal Area Networks (LR-WPANs).
- [38] IEC 62734 ed1.0 Industrial networks – wireless communication network and communication profiles – ISA 100.11a; 2014.
- [39] IEC 62591 ed1.0 Industrial communication networks – wireless communication network and communication profiles – WirelessHART™; 2010.
- [40] Liang W, Zhang X, Xiao Y, Wang F, Peng Z, Yu H. Survey and experiments of WIA-PA specification of industrial wireless network. *Wireless Commun Mobile Comput* 2011;11(8):1197–212.
- [41] Martinez J, Lastra JLM. Application of 6LoWPAN for the real-time positioning of manufacturing assets. Interconnecting Smart objects with the Internet; 2011. p. 3.
- [42] Kovatsch M, Duquennoy S, Dunkels A. A low-power CoAP for Contiki. In: Proceedings of 2011 IEEE 8th international conference on mobile adhoc and sensor systems (MASS). IEEE; 2011. p. 855–60.
- [43] Potsch T, Kuladinithi K, Becker M, Trenkamp P, Goerg C. Performance evaluation of CoAP using RPL and LPL in TinyOS. In: Proceedings of 5th international conference on new technologies, mobility and security (NTMS). IEEE; 2012.
- [44] OASIS Energy Market Information Exchange (EMIX) Version 1.0, January 2012. <<http://docs.oasis-open.org/emix/emix/v1.0/cs02/emix-v1.0-cs02.pdf>>.
- [45] IEC 61850-7-420 ed1 Communication networks and systems for power utility automation – Part 7-420: Basic communication structure – distributed energy resources logical nodes; 2009.
- [46] Luo Z, Alam M, Hong SH. A hardware-in-the-loop simulator for demand response energy management in industrial facilities. In: Proceedings of 2015



- IEEE workshop on modeling and simulation of cyber-physical energy systems (MSCPES); 2015.
- [47] Romkey JL. Nonstandard for transmission of IP datagrams over serial lines: SLIP. RFC 1988;1055.
- [48] Oikonomou G, Phillips I. Experiences from porting the Contiki operating system to a popular hardware platform. In: Proceedings of distributed computing in sensor systems and workshops (DCOSS), 2011 international conference on. IEEE; 2011.
- [49] Ancillotti E, Bruno R, Conti M. Reliable data delivery with the IETF routing protocol for low-power and lossy networks. *Ind Inform, IEEE Trans* 2014;10(3):1864–77.
- [50] Parolini L, Sinopoli B, Krogh BH, Wang Z. A cyber-physical systems approach to data center modeling and control for energy efficiency. *Proc IEEE* 2012;100(1):254–68.
- [51] Tham CK, Luo T. Sensing-driven energy purchasing in smart grid cyber-physical system. *Systems, Man, Cybernet: Syst, IEEE Trans* 2013;43(4):773–84.
- [52] Niemüller T, Ewert D, Reuter S, Karras U, Ferrein A, Jeschke S, Lakemeyer G. Towards benchmarking cyber-physical systems in factory automation scenarios. *Automation, Communication and Cybernetics in Science and Engineering* 2013/2014. Springer International Publishing; 2014. p. 665–9.